

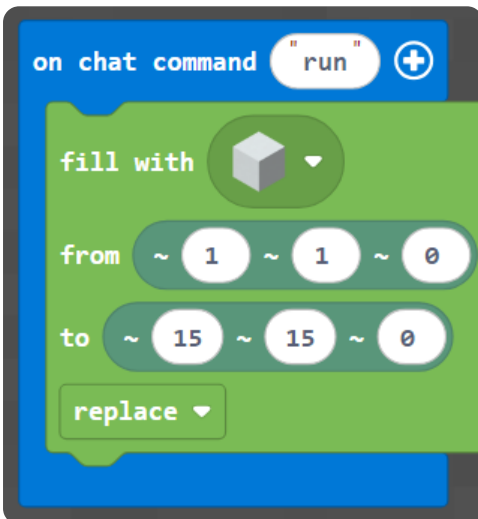
3D Coordinates Extension — English Worksheet (Intermediate)

Topic: Design Your Own Pixel Art (x, y) · **Course:** 3D Coordinates · **Level:** Intermediate (Extension, after Lesson 3) · **Time:** about 40 minutes

Every other week the picture was given to you. This lesson **you** draw it — then **you** read your own drawing back onto the wall.

Build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your home spot:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Stand on the red block every time you run.** If you move, the picture moves too.

1 · Predict 🌟

Read each set of steps. Write a sentence about what you will see.

```
place blue block at (3, 9)
place blue block at (4, 9)
place blue block at (5, 9)
```

Three blocks share $y = 9$. What shape do they make, and which number stays the same?


```
place blue block at (7, 2)
place blue block at (7, 3)
place blue block at (7, 4)
```

These three share $x = 7$. What shape do they make, and which number changes?


```
place green block at (6, 6)
place blue block at (6, 6)
```

Two blocks aim for the same (x, y) . What do you see — green, blue, or both? Why?

2 · Spot the Bug 🐛

Each one is broken. Fix it, then explain why the original was wrong.

Bug A — The flower's two leaves should sit at (2, 4) and (4, 4). Both land in the **same spot**.

```
place green block at (2, 4)
place green block at (2, 4)
```

Write the fixed line:

Why was it wrong?

Bug B — This should be a column going **up** at $x = 8$.

```
place black block at (8, 5)
place black block at (9, 5)
place black block at (10, 5)
```

Write the fixed lines:

Why was it wrong? Why does your fix work?

Stand on your **home spot** (the red block).

Your five rules.

#	Your design must...
1	fit inside the 15 × 15 wall
2	use 3 or more different block colours
3	use 25 or more blocks
4	be recognisable — someone else can name it without being told
5	be drawn on the grid below before you place a single block

Draw your design. Colour the squares. x goes **across**, y goes **up**.

 Your design

Count as you go. How many blocks did you use? How many colours?

Then build it. Read each coloured square's (x, y) and place that block on your wall.

Check your work. Hold your grid next to your wall. Is every block in the right spot? Write down any you had to move.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your grid. Hold this paper up to the camera.

2 • Your build. Hold your paper next to the wall so both are in the shot.

Fill the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was ____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.